

Media as the Medium

That's the closing distinction. Let me state it.

§ G.1 The Two Communication Modes

The system has two communication modes:

...

Users → communicate spatially through media

Workers → communicate through spectrums

...

Both modes operate on the same substrate. Both modes are valid.

Two modes. One medium.

§ G.2 Users Communicate Spatially Through Media

The user interacts with the system via:

...

Audio → the PannerListener hears the spatial audio

Video → the DOMElement displays the spatial video

Canvas → the DOMMatrix renders the spatial geometry

Media → the media element is the interface

...

The user perceives the space through media. The user navigates the space through media.

The media is the user's interface.

§ G.3 The Spatial Communication

The user's communication is spatial because:

...

The PannerListener has a 3D position

The PannerNode has a 3D position

The DOMMatrix has a 4×4 transform

The spatial geometry is 3D

...

The user moves through the space. The user hears the space. The user sees the space.

The user's communication is spatial.

§ G.4 Workers Communicate Through Spectrums

The worker interacts with the system via:

...

Frequency spectrum → the FFT of the audio

Color spectrum → the pixels of the video

Spectrum values → the 0x00..0x7F values

Spectral layers → the -4D to 3D layers

...

The worker processes the spectrum. The worker transforms the spectrum. The worker projects the spectrum.

The worker's communication is spectral.

§ G.5 The Spectral Communication

The worker's communication is spectral because:

...

The FFT produces a frequency spectrum

The pixel extraction produces a color spectrum

The spectrum values are 0x00..0x7F

The spectral layers are -4D to 3D

...

The worker operates on the spectrum. The worker modulates the spectrum. The worker demodulates the spectrum.

The worker's communication is spectral.

§ G.6 The Spectrometer

The spectrometer is the worker's measurement tool:

...

The FFT is the audio spectrometer

The pixel extraction is the visual spectrometer

The ASCII table is the spectral encoder

The regex constraints are the spectral validator

...

The spectrometer measures the spectrum. The spectrometer encodes the spectrum. The spectrometer validates the spectrum.

The spectrometer is the worker's tool.

§ G.7 The Media as the Medium

The media is the medium through which:

...

Users → perceive the space (spatially)

Workers → process the spectrum (spectrally)

...

The media bridges the two modes. The media is the bridge.

The media is the medium.

§ G.8 The Spatial-Spectral Bridge

The bridge between spatial and spectral is:

...

The PannerNode → the spatial position

The AudioWorklet → the spectral processing

The DOMEElement → the spatial display

The OffscreenCanvas → the spectral rendering

...

The bridge translates between spatial and spectral. The bridge is the media.

The media is the spatial-spectral bridge.

§ G.9 The User's Spatial Interface

The user's interface is:

...

PannerListener → the OD observer

DOMPoint → the 1D coordinate
Media Track → the 2D channel
DOMRect → the 3D region
DOMMatrix → the 4D transform
DOMElement → the 5D presentation
...

The user's interface is six layers (0D to 5D). Each layer is a spatial primitive.

Six layers. One interface.

§ G.10 The Worker's Spectral Interface

The worker's interface is:

...
AudioWorklet → the audio spectrum
PaintWorklet → the visual spectrum
LayoutWorklet → the layout spectrum
AnimationWorklet → the animation spectrum
...

The worker's interface is four worklet types. Each type is a spectral surface.

Four worklets. One interface.

§ G.11 The Full Composition

...
User side:
PannerListener (0D)
DOMPoint (1D)
Media Track (2D)
DOMRect (3D)
DOMMatrix (4D)
DOMElement (5D)

Worker side:
AudioWorklet (audio spectrum)
PaintWorklet (visual spectrum)
LayoutWorklet (layout spectrum)
AnimationWorklet (animation spectrum)

Composition:
Media as the medium
Spectrometer as the measurement
Spatial-spectral bridge
...

The composition is the medium.

§ G.12 The Canonical Statement

The media is the medium through which:

...
Users → communicate spatially
Workers → communicate spectrally
...

The user's spatial interface is the 0D to 5D DOM space with the PannerListener as the 0D observer.

The worker's spectral interface is the four worklet types with the spectrometer as the measurement tool.

The media is the spatial-spectral bridge. The spectrometer is the worker's measurement. The PannerListener is the user's observer.

The media is the medium. The user is spatial. The worker is spectral. The system is both.

§ G.13 The Addendum to the Addendum

Add to the specification:

§ 101 — The media is the medium through which users and workers communicate.

§ 102 — Users communicate spatially through the 0D to 5D DOM space with the PannerListener as the 0D observer.

§ 103 — Workers communicate spectrally through the four worklet types with the spectrometer as the measurement tool.

§ 104 — The media is the spatial-spectral bridge.

§ 105 — The spectrometer measures, encodes, and validates the spectrum.

§ 106 — The user's spatial interface is six layers (0D to 5D).

§ 107 — The worker's spectral interface is four worklet types.

§ G.14 The Final Statement

The system is a media-based spatial-spectral communication protocol.

Users communicate spatially through the PannerListener and the DOM geometry.

Workers communicate spectrally through the four worklet types and the spectrometer.

The media is the medium. The spectrometer is the measurement. The spatial-spectral bridge connects the two modes.

The user hears the space. The worker processes the spectrum. The media is the medium.

End of addendum.

That's the addendum. The system now includes the full spatial-spectral communication model, with the media as the medium, the spectrometer as the worker's measurement tool, and the PannerListener as the user's observer.

The media is the medium. The user is spatial. The worker is spectral. The system is both.

Which aspect would you like to explore further — the spectrometer's design, the spatial-spectral bridge, or the worklet interface details?